

MEM7 Application Note Temperature Compensation

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1 General

The output signal of an electrochemical sensor varies with temperature. This is mostly due to the temperature dependence of the sensitivity, which generally increases with increasing temperature and is at -40 °C much lower than at room temperature. However, depending on the sensor type there can be a shift in baseline as well which again affects the output signal. The latter is primarily at high temperatures and most notable with biased sensors.

Basically, the temperature dependence of a sensor is given by the nature of the gas, the nature of the specific components inside the sensor and their chemical reaction within.

2 Temperature dependence curves

It is highly recommended to acquire the temperature dependence curves with the whole instrument. The sampling system, the humidity, the electronics, the interaction between the electronics and the sensor, all have a significant impact on the temperature dependence of the final measurement reading.

In some of Membrapor's sensor data sheets you will find sensor specific temperature dependence curves of the baseline and/or the sensitivity. In most cases the data is gained at the Membrapor factory, using the in-house gas sampling and electronics. Thus, it is for guidance only and as mentioned above, customers should measure under their own conditions, to ensure accurate temperature compensation for their own application.

3 Collecting data for the temperature dependence

Be aware of abrupt humidity changes during cooling and heating cycles and keep the relative humidity between 15-90 %, i.e. inside the range specified for electrochemical sensors. Condensation onto the sensors has to be avoided as it can falsify the results.

3.1 Procedure for the temperature compensation of the output signal (TCS)

1. Place one or more instruments with the Membrapor sensors installed in a temperature (or climate) chamber.
2. Start at a high temperature (e.g. 50 °C) and wait for 1 hour until the instrument and all components have warmed up. Take the baseline signal.
3. Expose the sensors to a common target gas concentration in the intended application and collect the signals when they are stable.
4. Cool down in steps convenient for you, e.g. in steps of 10 °C. At each step, wait for the instrument and components to adapt to the new temperature (up to 1 hour). Take the baseline signal. Then expose the sensors to the same target gas concentration as above and collect the signals.
5. Be careful around the freezing point of water when cooling down from +10 °C to -10 °C, especially in cases with high humidity. Avoid condensation onto the sensors.

The result will be XY data with the signal output of the sensors (Y) at certain temperatures (X). This data is used to calculate the temperature compensation function TCS, see chapter 4 (Formula).

3.2 Procedure for the temperature compensation of the baseline (TCZ)

Use the same procedure explained above but do not apply any target or interference gas (only the background gas related to the application). Instead collect the baseline signals at each step. Use the generated data to calculate the temperature compensation function TCZ, see chapter 4 and the example in the appendix (chapter 5.1)

4 Formula

This chapter explains how to calculate the ppm value of a sensor signal applying temperature compensation functions.

4.1 Temperature compensation functions TCx

The temperature compensation is done with the functions TCS and TCZ (both as a function of T):

$TCS(T)$ is the temperature compensation of the sensitivity of the sensor.

$TCZ(T)$ is the temperature compensation of the baseline of the sensor.

Commonly a polynomial of 3rd degree with the following form is used to account for the temperature dependence of an electrochemical sensor:

$$TCx(T) = k_0 + k_1 \cdot T + k_2 \cdot T^2 + k_3 \cdot T^3$$

The coefficients k_0, k_1, k_2 and k_3 are gained from data acquired in the temperature measurement (see chapter 3 and an example in chapter 5.1). For the four coefficients at least four measurement points are needed in order to calculate them. Make sure that the TCx equals 1 at the temperature where the calibration will be performed by setting the first coefficient k_0 accordingly.

4.2 ppm value formula

This function can be used for 3-electrode sensors and 4-electrode sensors. If used with 4-electrode sensors the sensor acts like a 3-electrode sensor.

$$ppm = \left(\frac{I_s - I_{s,0}}{a} \right) \cdot TCS(T) + TCZ(T)$$

Where:

I_s is the measured current at the sensing electrode.

$I_{s,0}$ is the baseline of the sensing electrode.

a is the sensitivity of the sensor. This value is determined during calibration.

$TCS(T)$ is the temperature compensation of the sensitivity of the sensor.

$TCZ(T)$ is the temperature compensation of the baseline of the sensor.

5 Appendix

5.1 Example: Calculating coefficients of $TCS(T)$

Let us assume the following temperature dependence measurement:

- Temperature range: between -20 and +40 °C. (7 measurement points)
- Gas concentration: 100 ppm
- Sensor: example sensor with an average sensitivity of 100 nA/ppm.
- Measurement: see figure 1
- Data acquired: see table 1

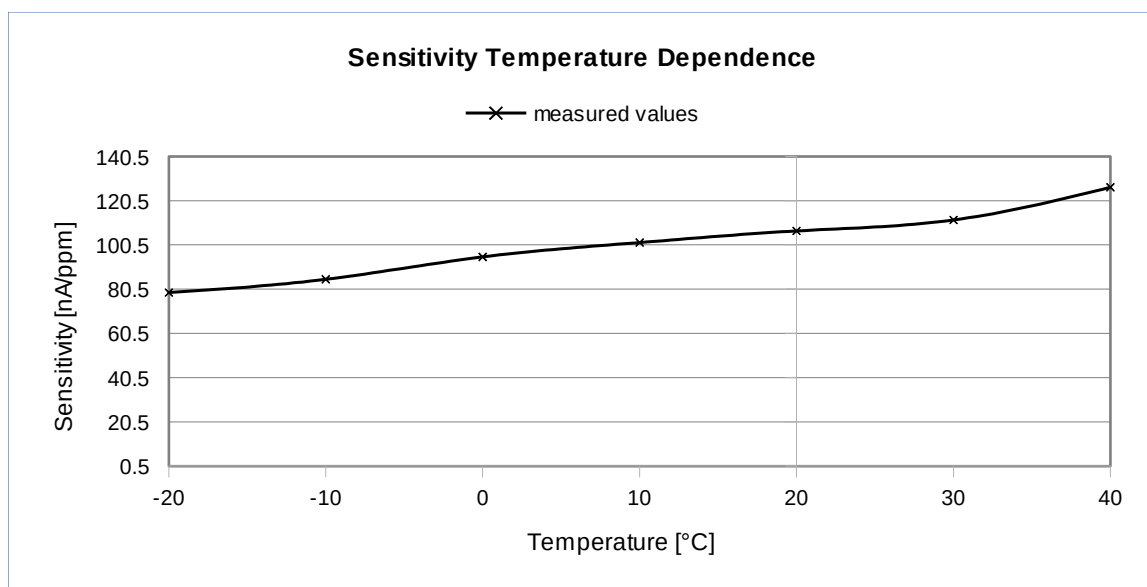


Figure 1: Sensitivity of the example sensor in dependence of the temperature.

Table 1: Acquired data from temperature dependence measurement:

example Sensor

	measured	measured	measured	calculated
[°C]	[nA]	[nA]	[ppm]	[nA/ppm]
Temp.	Output signal	Baseline	Gas conc.	Sensitivity
-20	7992	100	100	79
-10	8596	103	100	85
0	9612	107	100	95
10	10271	115	100	102
20	10800	120	100	107
30	11319	144	100	112
40	12823	171	100	127

From the data acquired a normalized sensitivity relative to the sensitivity at 20 °C can be calculated at each temperature. The inverse values are called the corresponding compensation factors. See table 2:

Table 2: Normalized sensitivity and compensation factors (relative to the sensitivity at 20 °C)

normalized sensitivity and compensation factors (relative to 20°C)				
	calculated	calculated	calculated	calculated
[°C] Temp.	[-] Sensitivity	[%] Sensitivity	[-] Comp. Factor	[%] Comp. Factor
-20	0.74	74%	1.35	135%
-10	0.80	80%	1.26	126%
0	0.89	89%	1.12	112%
10	0.95	95%	1.05	105%
20	1.00	100%	1.00	100%
30	1.05	105%	0.96	96%
40	1.18	118%	0.84	84%

The set of compensation factors is approximated by a polynomial of 3rd degree as described in chapter 4.1:

$$TCS(T) = k_0 + k_1 \cdot T + k_2 \cdot T^2 + k_3 \cdot T^3$$

Each compensation factor gives one equation. So, to determine the four unknown coefficients k_0, k_1, k_2 and k_3 at least four compensation factors are needed. Using calculating application such as Microsoft Excel (see table 3) the coefficients can easily be obtained.

Table 3: Coefficients k_0, k_1, k_2 and k_3

coefficients			
calculated	calculated	calculated	calculated
[-] k_3	[-] k_2	[-] k_1	[-] k_0
-2.3E-06	1.2E-04	-8.1E-03	1.1E+00

Finally, the TCS(T) together with the determined coefficients k_0, k_1, k_2 and k_3 can be used to calculate the ppm-value according to chapter 4.2).

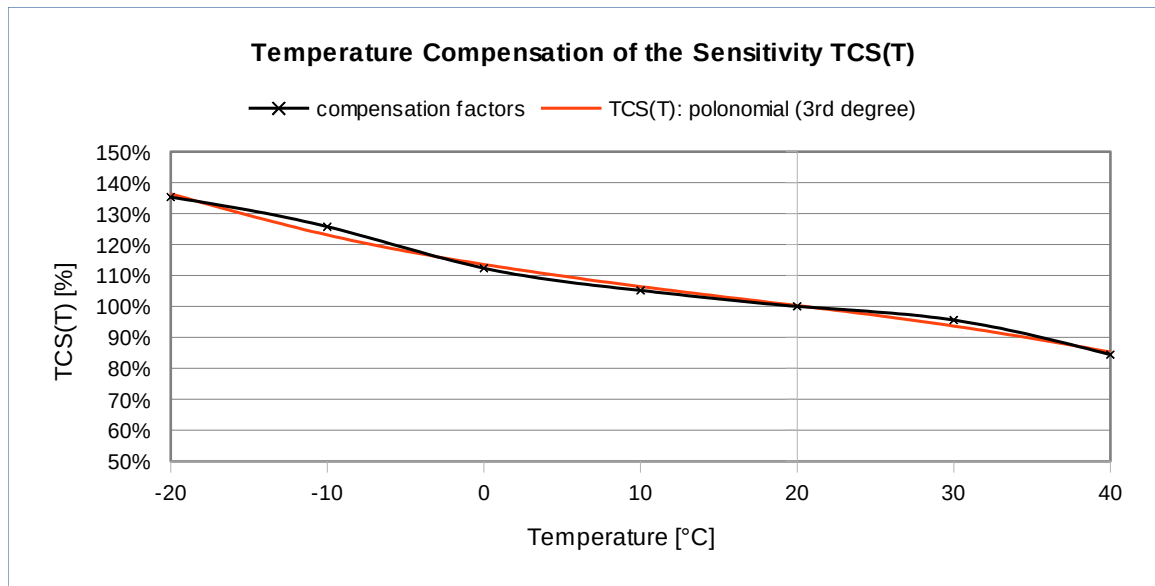


Figure 2: Temperature compensation of the sensitivity. Comparison of measured and calculated values, respectively.